Quantitative Project Management

MGT 207

10/8/26

Why Project Management is Core to Operations



"Run" vs. "Change" the Business

- **Run the Business (Steady-State):** Exploitation, efficiency, variance reduction, executing existing capabilities reliably.
- **Change the Business (Projects):** Exploration, new capabilities, strategic transformation.

"Projects build the processes that Operations later runs."

Shared Analytical DNA

Repetitive OM	↔ **Project OM**
Bottlenecks (TOC)	↔ **Critical Path (CPM)**
Line Balancing	↔ **Resource Leveling**
Expediting / Overtime	↔ **Project Crashing**
Safety Stock / Pooling	↔ **PERT & Time Buffers**

Same physics of flow and variability—applied to a lifecycle of one.

Why Project Management Matters More Than Ever

Flyvbjerg's "Iron Law" of Megaprojects

99.5%

of major projects fail to deliver **On-Time**, **On-Budget**, and **On-Benefits**

(Flyvbjerg, 2023 — $N > 16{,}000$ projects)

- **IT / ERP:** $>$ 400% tail-event cost overruns
- **Aero & Infra:** Multi-year integration delays
- **Tech / R&D:** Late launch destroys ROI

How Project Management Creates Value Today

- **Knowing Where to Spend Money to Save Time (Crashing):** When a project falls behind, CPM tells you *which* bottleneck tasks are worth paying extra to speed up—and which tasks have slack and don't matter.
- **Avoiding the "Panic-Hiring" Trap (Brooks's Law):** Team communication links grow as $\frac{n(n-1)}{2}$ (5 people = 10 links; 15 people = 105 links). Adding staff to a late project often makes it *later* due to training and meeting overload.
- **Why "Agile" Doesn't Replace CPM:** Agile sprints are great for software and prototyping, but physical supply chains, factories, and product launches still have hard sequential dependencies that require critical-path planning.

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  - Helpful even when the details don't unfold as expected
  - Eisenhower: "The plan is nothing. Planning is everything."

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  - Who do we go to for help making decisions?
  - How do we check progress on a known issue?

Overview

I. **Project scheduling with known activity times**
   - Example
   - CPM approach

II. **Project Crashing**
   - Heuristic approach

III. **Project scheduling with uncertain activity times**
   - PERT approach
   - Simulation approach

IV. **Case study discussion**

Product Launch at AeroTech Solutions

AeroTech Solutions has designed a revolutionary new autonomous delivery drone targeted at commercial logistics companies. R&D and core design are complete. All that remains is to plan the manufacturing ramp-up and the global marketing campaign. The company has identified 9 key activities and the estimated costs for the project launch. How should AeroTech schedule these activities to hit the market in the shortest possible time?

Activity	Activity Description	Immediate Predecessor	Activity Time (weeks)	Normal Cost
A	Conduct market analysis	–	6	$240,000
B	Secure manufacturing facilities	A	4	$300,000
C	Hire production managers	A	3	$150,000
D	Procure raw materials	B	3	$540,000
E	Hire & train assembly team	B, C	10	$900,000
F	Manufacture pilot units (prototypes)	D, E	2	$300,000
G	Scale to full production	F	6	$1,350,000
H	Develop marketing strategy	B	6	$450,000
I	Execute global launch campaign	H, F	8	$1,050,000
		Total	**48**	

Construction of a Network Diagram

Given:

1. list of all activities
2. precedence relationships
3. estimates of time and cost for each activity

We are able to construct a network diagram where:

1. each node represents an activity
2. each arc represents the precedence relationship between two activities
3. there is a (dummy) starting node ST and a (dummy) finishing node FIN

Scheduling Calculations

After creating a network of a project, the next step in the CPM is to determine the earliest time that each activity in the network can start and finish. We determine these by what is called a forward pass.

A. Notation

- t_k = estimate of the mean duration time of activity k
- ES_k = earliest start time for activity k
- EF_k = earliest finish time for activity k
- LS_k = latest allowable start time for activity k
- LF_k = latest allowable finish time for activity k

ES_k	EF_k
k	t_k
LS_k	LF_k

Forward Pass

B. Rules

(1) Forward Pass (ES, EF)

Step 1: for START node, set $ES_{start} = 0$

Step 2: an arbitrary activity k is assumed to start as soon as all of its predecessor activities are completed. This may be written as:

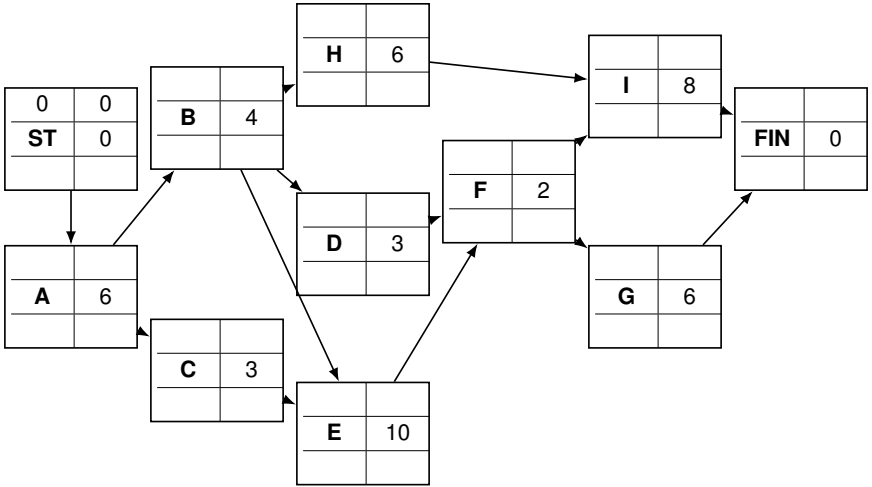
$$ES_k = \text{largest EF of activity } k\text{'s predecessors} = \max_{j \in k\text{'s immed. pred.}} \{EF_j\}$$

$$= \text{the earliest possible time to start activity } k$$

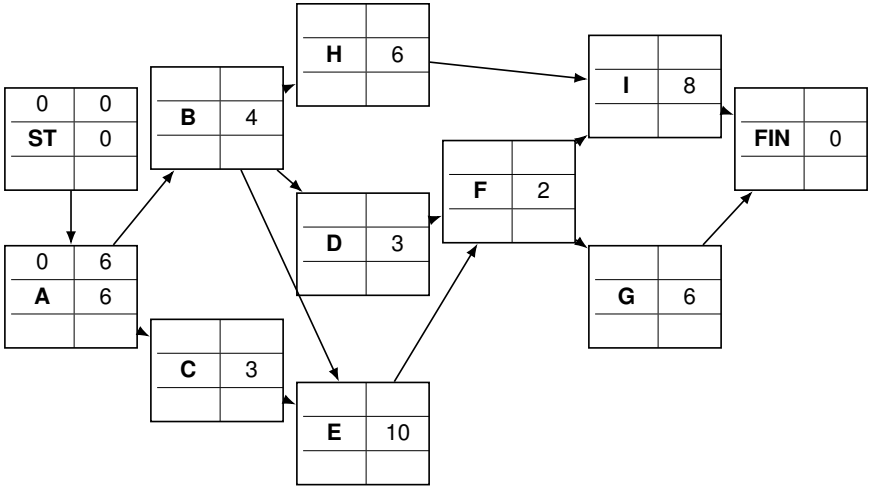
Step 3: the earliest possible finish time of activity k is the sum of its earliest start time and the estimated activity time.

$$EF_k = ES_k + t_k$$

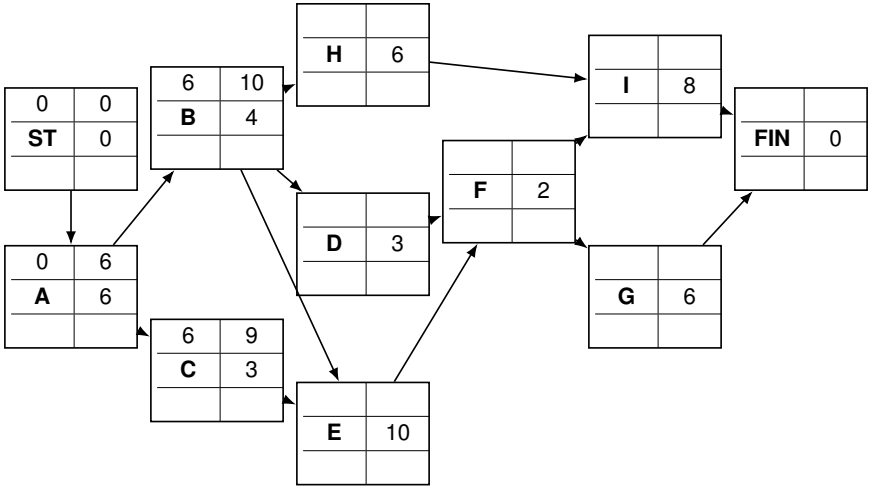
Forward Pass (Step-by-Step)



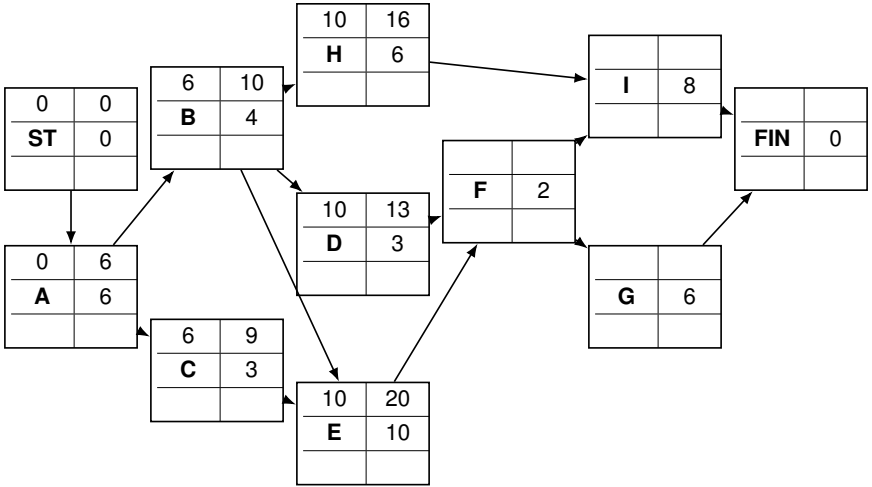
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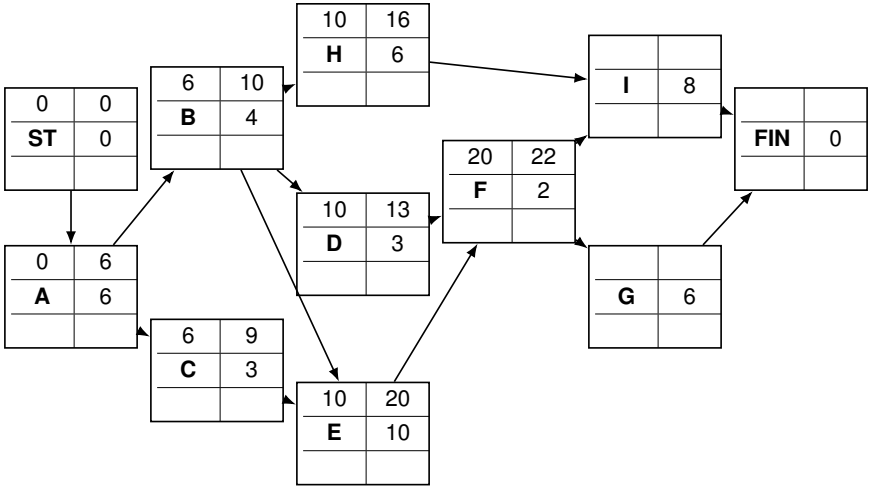
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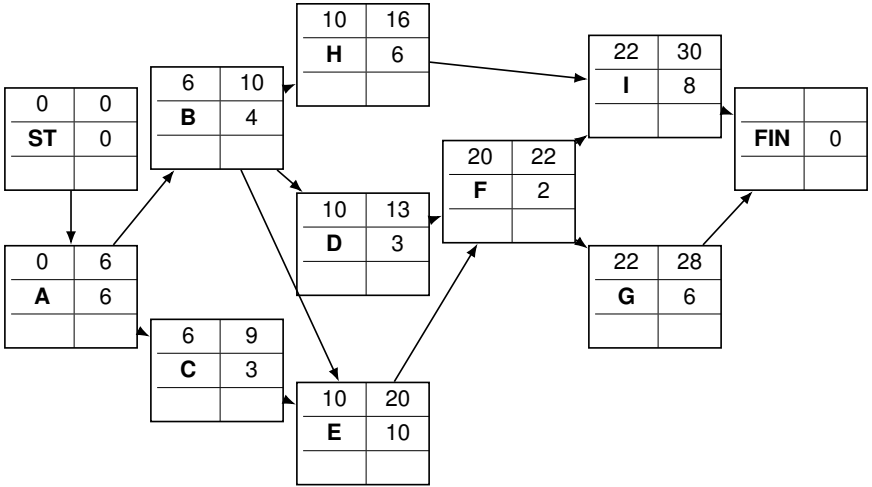
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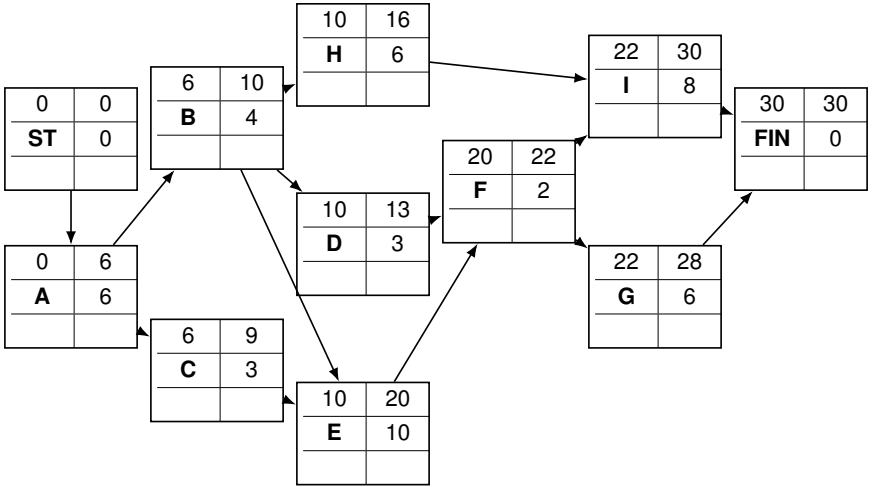
Forward Pass (Step-by-Step)



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Backward Pass

(2) Backward Pass (LS, LF)

Step 1: LF of FINISH node is set equal to its earliest finish time as computed in the forward pass computations: $LF_{FIN} = EF_{FIN}$

Step 2: the latest allowable finish time for activity k without delaying the entire project is the smallest (or earliest) of the latest allowable start times of its successor activities.

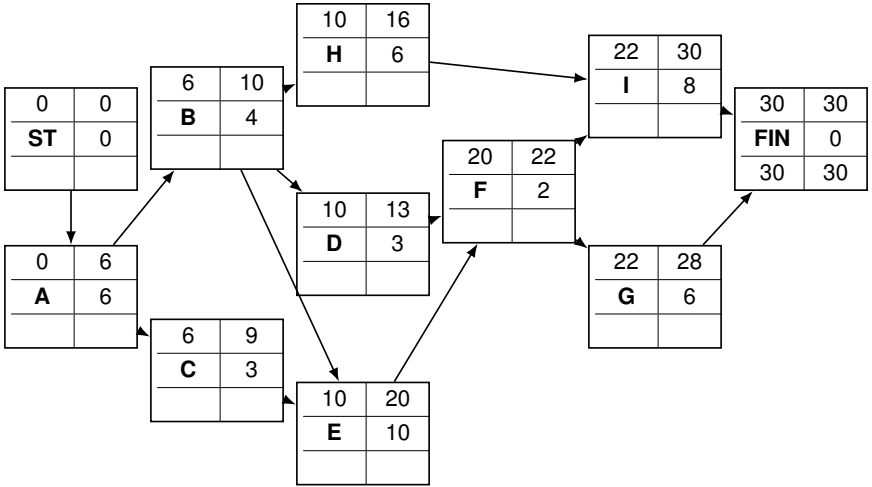
$$LF_k = \text{smallest LS of } k\text{'s successors} = \min_{j \in k\text{'s immed. succ.}} \{LS_j\}$$

$$= \text{latest allowable finish time for activity } k \text{ without affecting proj. completion}$$

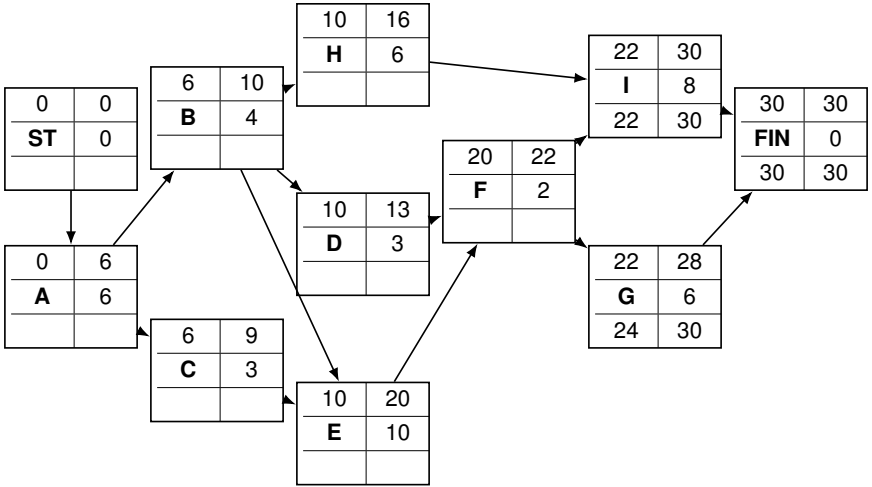
Step 3: the latest allowable start time for activity k without delaying the entire project is its latest allowable finish time minus the estimated activity time.

$$LS_k = LF_k - t_k$$

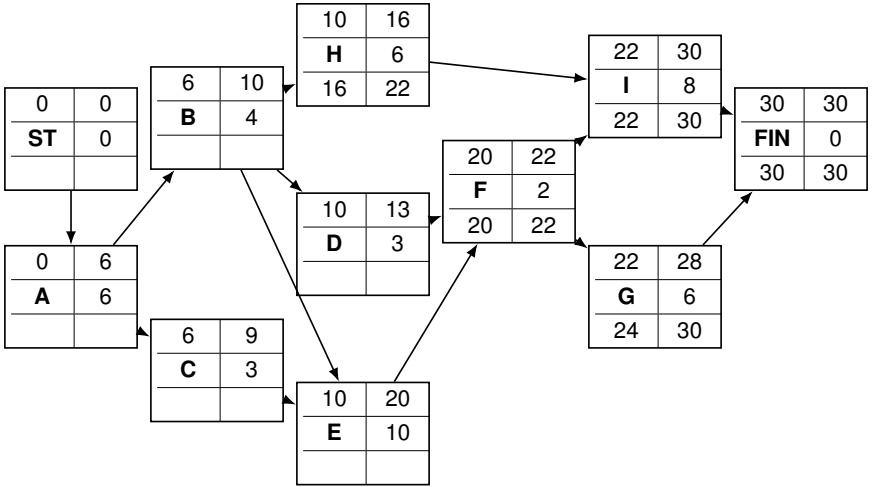
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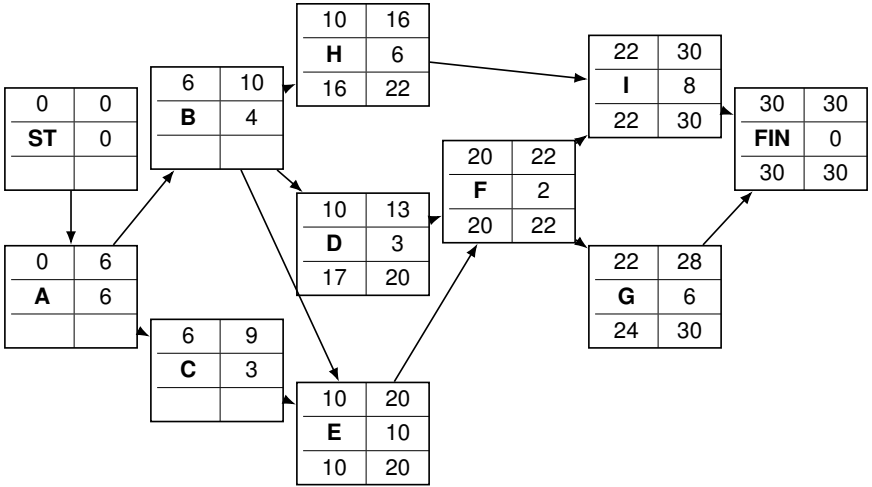
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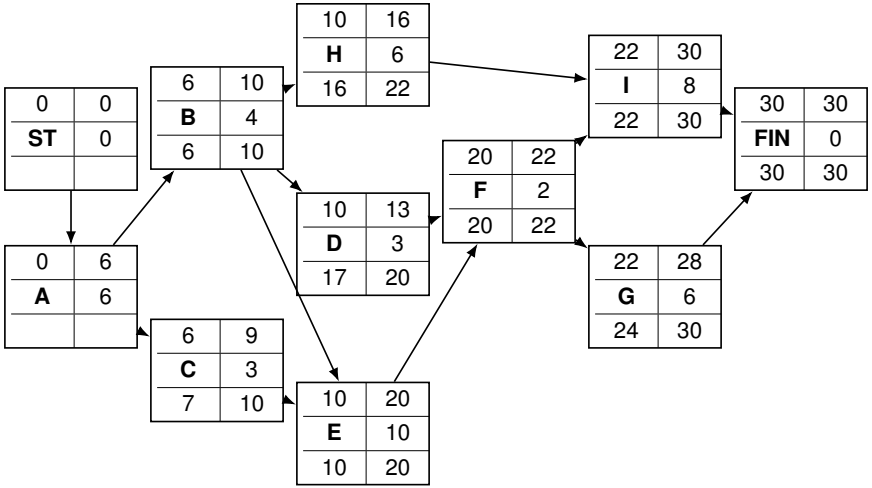
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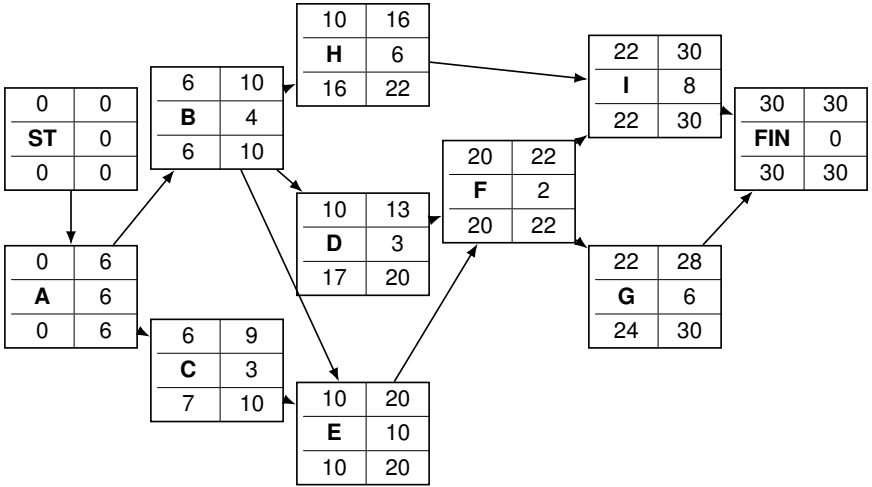
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After Backward Pass: Slack

We know:

- The latest allowable starting time and the latest allowable finishing time for each activity

So... Define Slack:

$$SLACK_k = LS_k - ES_k = LF_k - EF_k$$

= the length of time an activity can be delayed without affecting the completion date for the entire project, assuming all its immediate predecessors start at ES.

i.e., activity k can be delayed by $SLACK_k$ from its ES_k without affecting the project completion time.

Connection to the Harvard Case Simulation

Speeding up a task with positive slack (e.g., by adding resources or overtime) **wastes budget without shortening the overall project schedule**. You must identify the Critical Path first!

Critical Path

- **Path**

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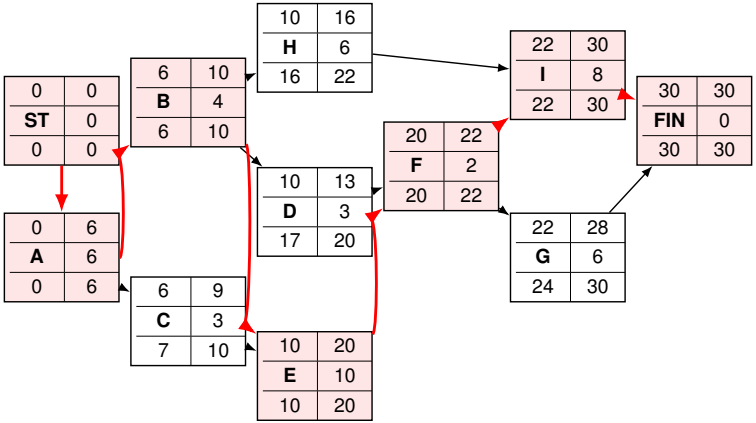
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- Clearly, critical activities are those that have **zero slack**...

CPM: Critical Path Identified



Critical Path: Start - A - B - E - F - I - Finish

Time-Cost Tradeoffs – Crashing Activities

The management of AeroTech Solutions is giving some thoughts to putting extra resources into the project so that it can be completed within **26 weeks**. Each activity has been studied, and a set of crash times (in weeks) and cost has been developed. Determine a crash plan which minimizes the total cost of completing the project within 26 we

Connection to the Harvard Case Simulation

When you decide to authorize overtime, hire more staff, or outsource to hit a looming deadline, you are effectively **"crashing"** the project. You must weigh the **cost per week saved**.

Activity	Activity Description	Normal Time	Normal Cost	Crash Time	Crash Cost
A	Conduct market analysis	6	$240,000	5	$300,000
B	Secure manufacturing facilities	4	$300,000	3	$400,000
...	...	...	...	...	...

A Heuristic Approach

What activities to crash and how much to crash?

Act.	Time (T_i)	Budgeted Cost (C_i)	Crash Cost (C_i')	ΔCost ($C_i' - C_i$)	ΔTime ($T_i - T_i'$)	Crash Cost / Wk ($\Delta C/\Delta T$)	Slack (From CPM)
A	6	$240,000	$300,000	$60,000	1	$60,000	0
B	4	$300,000	$400,000	$100,000	1	$100,000	0
C	3	$150,000	$240,000	$90,000	1	$90,000	1
D	3	$540,000	$750,000	$210,000	1	$210,000	7
E	10	$900,000	$1,440,000	$540,000	3	$180,000	0
F	2	$300,000	$390,000	$90,000	1	$90,000	0
G	6	$1,350,000	$3,200,000	$1,850,000	2	$925,000	2
H	6	$450,000	$900,000	$450,000	3	$150,000	6
I	8	$1,050,000	$2,600,000	$1,550,000	4	$387,500	0

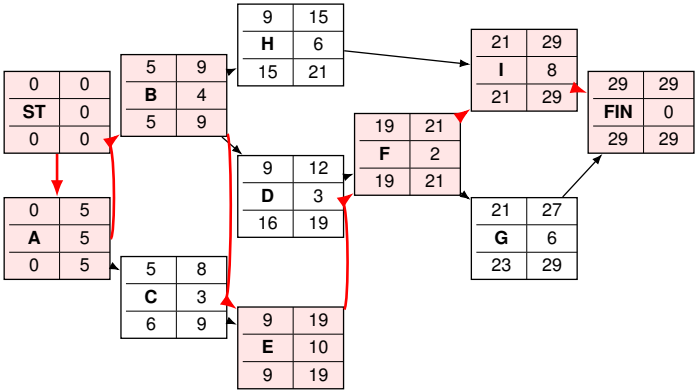
Current Critical Path: A - B - E - F - I (30 weeks)
Need completion in 26 weeks.

Crashing Step 1: Crash Activity A by 1 Week

- Activity A is on the critical path and has the lowest Crash Cost / Week ($60,000).
- New Duration for A = 5.
- Entire project duration drops to **29 weeks**.

Next: Which activity to crash next?

- Check critical path. It remains **A-B-E-F-I**.
- Next cheapest on CP is F ($90k/wk).

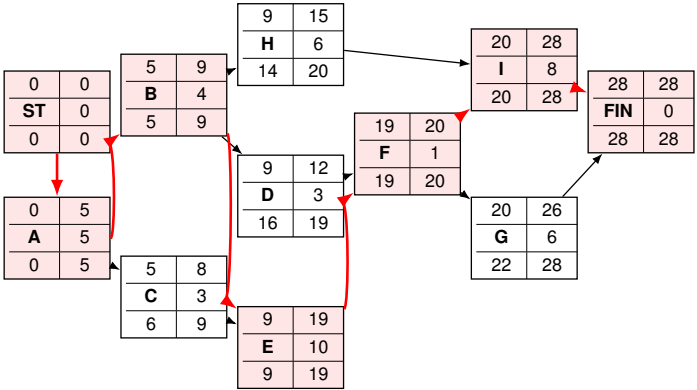


Crashing Step 2: Crash Activity F by 1 Week

- Crash F by 1 week. Cost = $90,000.
- New Duration for F = 1.
- Entire project duration drops to **28 weeks**.

Next: Which activity to crash next?

- Check critical path. It remains **A-B-E-F-I**.
- Next cheapest on CP is B ($100k/wk).

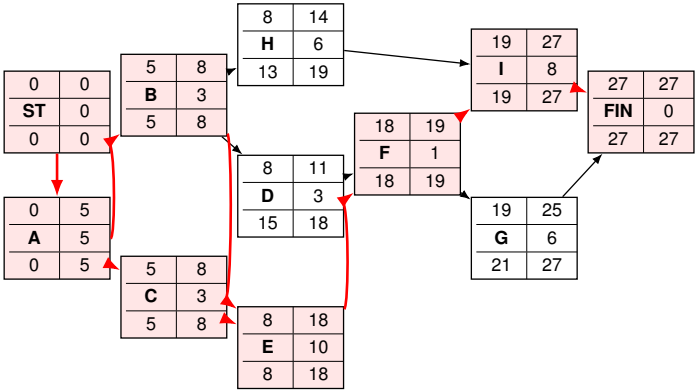


Crashing Step 3: Crash Activity B by 1 Week

- Crash B by 1 week. Cost = $100,000.
- New Duration for B = 3.
- Entire project duration drops to **27 weeks**.

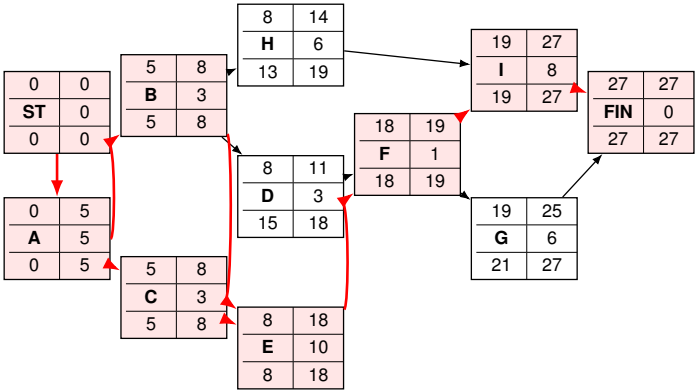
Next: Which activity to crash next?

- Check critical path. It is STILL A-B-E-F-I.
- But wait, Activity C's slack has dropped to 0!
- We now have **two** critical paths: **A-B-E-F-I** and **A-C-E-F-I**.



Crashing Step 4: What if we crash C?

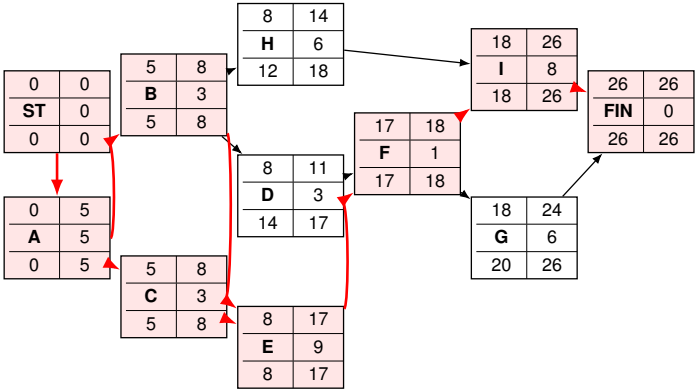
- If we crash C (cost $90,000/wk)...
- **Doesn't help.** Because activity E cannot start until *both* B and C finish.
- B is 3 weeks, C is 2 weeks. E still waits for B. The total duration remains 27.
- To save time, we would have to crash *both* B and C together (cost $190,000/wk).



Crashing Step 5: Crash Activity E by 1 Week

- Instead of crashing B and C, crash E.
- E is on *both* critical paths!
- Crash E by 1 week. Cost = $180,000.
- New Duration for E = 9.
- Entire project duration drops to **26 weeks**!

Success! We achieved 26 weeks.



About Heuristic Approach

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- **Remember:**
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- **Recommendation:**
  - Crash activities one by one.
  - After crashing one activity, check the critical path(s) in the revised network.

Project Scheduling with Uncertain Activity Times

The R&D team at AeroTech used a completely new AI navigation chip for the drone. Given the massive market hype, management wants to roll out the product as soon as possible. However, due to supply chain complexities and new manufacturing techniques, the operations managers can only give a rough estimate (optimistic, most probable, pessimistic) on the duration time of activities. **What would be the expected time required to complete the project?**

Connection to the Harvard Case Simulation

In the game, you will face unexpected curveballs (e.g., staff quitting, tech failures). PERT proves mathematically why a single deterministic plan is an illusion, and why you must manage risk dynamically.

Activity	**Optimistic (a_i)**	**Most Probable (m_i)**	**Pessimistic (b_i)**
A	4	6	7
B	4	6	9
...	...	...	...

Three-Point Estimation Technique (PERT)

A. Time estimates

1. For each activity, estimate:
   - a_i = optimistic time
   - b_i = pessimistic time
   - m_i = most probable time
2. Compute approximate value assuming a beta probability distribution:

$$\text{Expected Time} = \mu_i = \frac{a_i + 4m_i + b_i}{6}$$

$$\text{Variance} = \sigma_i^2 = \left(\frac{b_i - a_i}{6}\right)^2$$

Critical Path & Probabilistic Analysis

B. Procedure

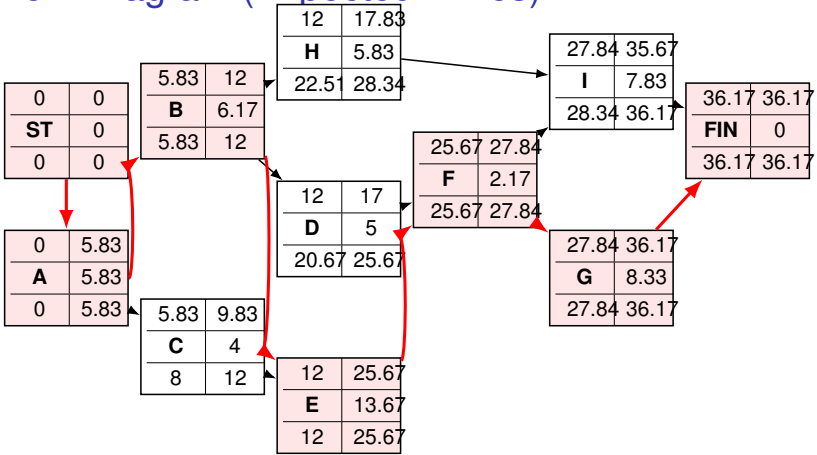
1. Treat the expected activity times as the known duration of activity. Use the PERT/CPM procedure to find the critical path of the project.
2. Compute estimates of mean and variance of project completion time as sum of expected times and variances along critical path:
   - Mean (μ) = Sum of Expected Times on Critical Path
   - Variance (σ^2) = Sum of Variances on Critical Path
3. Assume the project completion time is normally distributed $N(\mu, \sigma)$
4. Use Standardized Normal Distribution Table to determine certainty of completing project within a given time frame.

AeroTech PERT Calculations

	Activity	Optimistic	Most Prob.	Pessimistic	Mean Time (μ_i)	Variance (σ_i^2)
✓	A	4	6	7	5.83	0.25
✓	B	4	6	9	6.17	0.69
	C	3	4	5	4	0.11
	D	3	5	7	5	0.44
✓	E	10	14	16	13.67	1
✓	F	2	2	3	2.17	0.03
✓	G	6	8	12	8.33	1
	H	4	6	7	5.83	0.25
	I	6	8	9	7.83	0.25

CPM gives Critical Path: A-B-E-F-G

PERT Network Diagram (Expected Times)



Critical Path: Start - A - B - E - F - G - Finish

Project Completion Probabilities

Suppose the critical path is: A - B - E - F - G

(1) What is the estimated expected completion time?

- $\mu_{CP} = 5.83 + 6.17 + 13.67 + 2.17 + 8.33 = 36.17$
- $\sigma^2_{CP} = 0.25 + 0.69 + 1 + 0.03 + 1 = 2.97$

(2) What is the probability of completing the project in 33 weeks?

- `NORM.DIST(33, 36.17, SQRT(2.97), 1) = 0.0329 or 3.3%`

(3) With 90% probability, what is the project completion time?

- `NORM.INV(0.9, 36.17, SQRT(2.97)) = 38.37 days`

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- **The Human Element:** You must also manage team morale, stress, and coordination meetings (standups, coaching).

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- **Apply the Class Tools:** How can identifying **slack** from CPM help you optimize resources in this scenario?

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Scenario C: Mid-Project Surprise

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Scenario C: Mid-Project Surprise

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- **Apply the Class Tools:** How is this like **crashing** activities after you have already started the project? Did any new activities suddenly become critical?

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Let's log in and play!